



**JAWAHARLAL COLLEGE
OF ENGINEERING & TECHNOLOGY**
JAWAHAR GARDENS, LAKKIDI MANGALAM, PALAKKAD
(An ISO 9001:2015 Certified Institution)



DEPARTMENT OF COMPUTER SCIENCE (CYBERSECURITY)

ZERO-TRUST SECURITY FOR MODERN NETWORKS

ABSTRACT

Traditional network security assumes that users and devices inside an organization's network can be trusted. However, with the growth of cloud computing, remote work, mobile devices, and distributed applications, this approach has become increasingly difficult to maintain. Zero-Trust Security is a modern cybersecurity approach based on the principle of "never trust, always verify." This seminar explores how Zero-Trust architecture can protect organizational resources by continuously verifying users, devices, applications, and access requests regardless of their network location. The concept includes strong identity verification, multi-factor authentication, least-privilege access, device security, network segmentation, and continuous monitoring. Unlike traditional perimeter-based security, Zero Trust assumes that threats may already exist inside the network and therefore limits access to only the resources required by an authorized user or application. The seminar also discusses the advantages and challenges of implementing Zero-Trust security in organizations, including integration with existing systems, user management, and operational complexity. By examining real-world applications and security principles, this topic demonstrates how Zero-Trust architecture can reduce unauthorized access, limit the impact of compromised accounts, and provide stronger protection for modern digital environments.

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